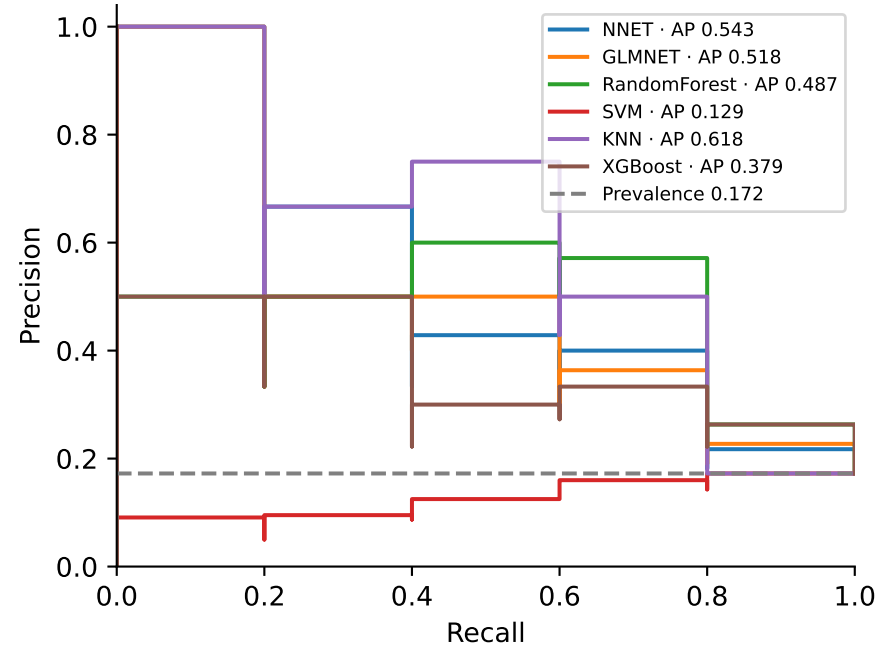
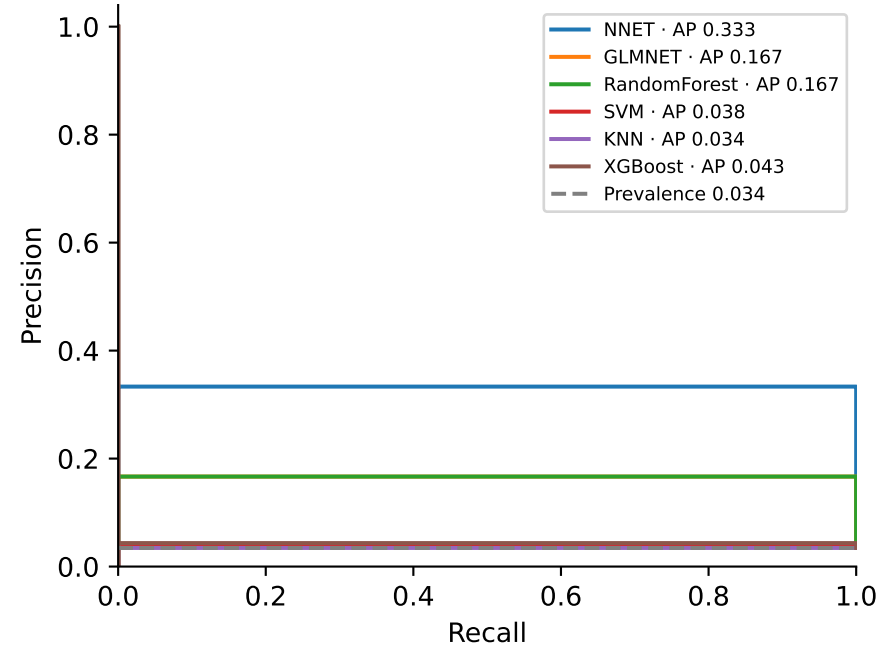


Precision-recall comparison across all six models

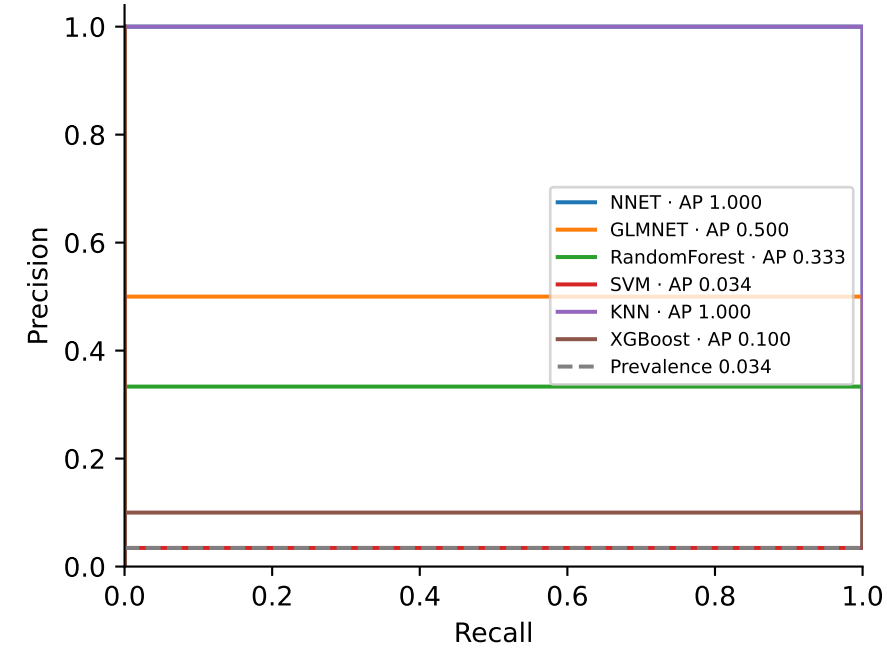
Acid-base balance · 5 positive



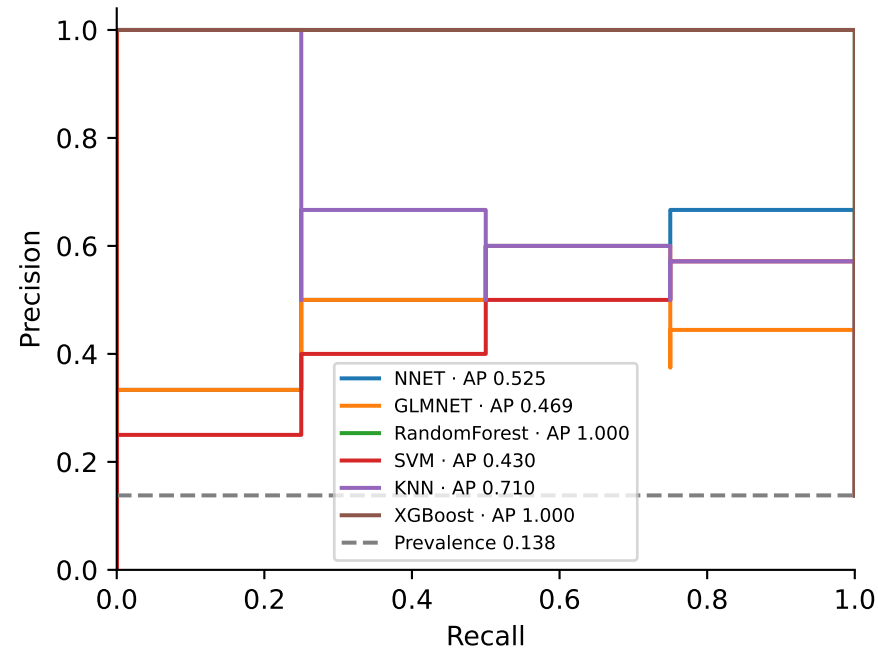
Buffer capacity · 1 positive



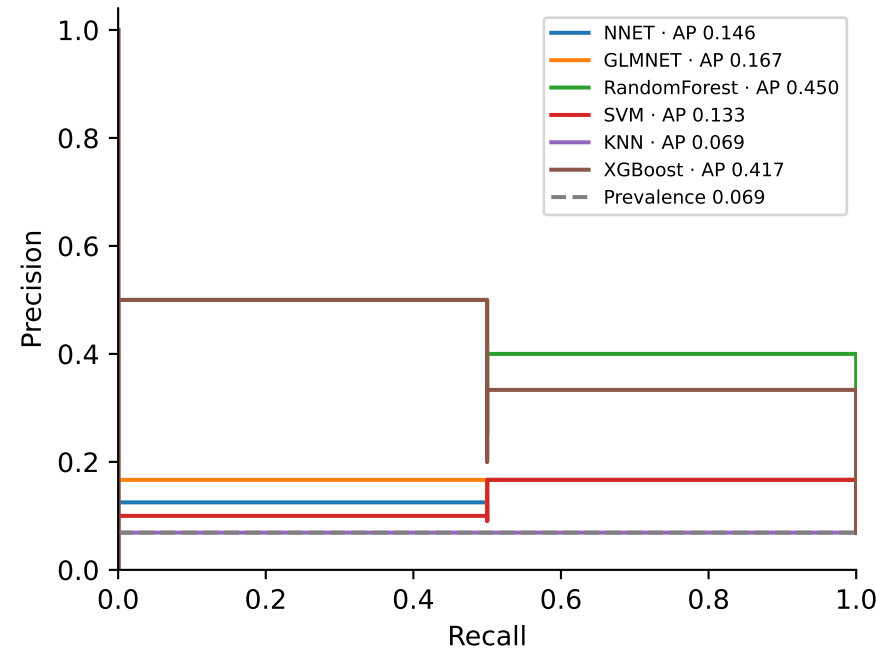
Acid accumulation · 1 positive



Ammonia toxicity · 4 positive



Biogas quality · 2 positive



Independent test set

Curves use predicted probabilities.
AUPR means average precision.

One-positive curves are unstable;
read them alongside label support.

No test-based threshold tuning.